

Fake Review Detection

Final Project Report

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Abstract

The rapid growth of e-commerce platforms has made online customer reviews one of the most influential factors in purchasing decisions. However, the increasing presence of fake reviews reduces consumer trust and makes it more difficult to distinguish between authentic and misleading opinions. As a result, automatic fake review detection has become an important Natural Language Processing (NLP) task.

This project presents a complete machine learning pipeline for detecting fake product reviews using a publicly available dataset containing 40,432 Amazon reviews. The project includes exploratory data analysis, text preprocessing, TF-IDF feature extraction, model comparison, hyperparameter tuning, and deployment of the final model as an interactive Streamlit web application.

Several classical machine learning algorithms were evaluated, including Multinomial Naive Bayes, Logistic Regression, and Linear Support Vector Machine (Linear SVM). Additional improvements such as bigram features, GridSearchCV optimization, and probability calibration were applied to improve the model performance.

The final calibrated Linear SVM achieved an accuracy of approximately **90.17%**, outperforming all previous models while providing confidence scores for each prediction. The resulting application allows users to enter a product review and instantly receive both the predicted class and the model confidence.

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1 Introduction

Online product reviews have become one of the primary sources of information for consumers when making purchasing decisions. Positive reviews can significantly increase product sales, while negative reviews may discourage potential buyers. Because of their influence, fake reviews have become a widespread problem across many e-commerce platforms.

Fake reviews are often generated to artificially improve or damage the reputation of products, services, or sellers. Such reviews reduce consumer trust and make it more difficult for users to identify genuine customer experiences. As online marketplaces continue to grow, automatically detecting fake reviews has become an important application of machine learning and Natural Language Processing (NLP).

In this project, we developed a machine learning pipeline for classifying product reviews as **Fake** or **Real** using a publicly available dataset. Several classical machine learning algorithms were trained and compared in order to identify the most effective model for this task. Throughout the project, different feature engineering techniques and model optimization methods were evaluated to improve the overall classification performance.

The final solution was deployed as an interactive Streamlit application, allowing users to enter any product review and receive an immediate prediction together with the model's confidence score.

2 Dataset

The dataset used in this project was obtained from Kaggle and contains **40,432 Amazon product reviews** collected from multiple product categories.

The dataset is balanced and consists of two classes:

OR (Original Reviews) – authentic human-written reviews.

CG (Computer-Generated Reviews) – automatically generated fake reviews.

Using a balanced dataset is beneficial because it prevents the model from becoming biased toward one class and allows accuracy to provide a meaningful measure of performance.

The reviews cover a variety of product categories, including books, electronics, kitchen products, movies, apparel, sports equipment, health products, and additional consumer goods. This diversity makes the classification task more representative of real-world review data.

Before training any model, the dataset was explored through descriptive statistics and several visualizations to better understand its characteristics.

2.1 Exploratory Data Analysis

Before training the machine learning models, an exploratory data analysis (EDA) was performed to better understand the dataset and identify important characteristics of the reviews. The analysis focused on the class distribution, rating distribution, product categories, review length, word count, and the most common words appearing in fake and real reviews.

The first step was to verify the distribution of the target labels. As shown in **Figure 1**, the dataset is almost perfectly balanced between fake (CG) and real (OR) reviews. This balanced distribution is beneficial because it prevents the classifier from becoming biased toward one class during training.

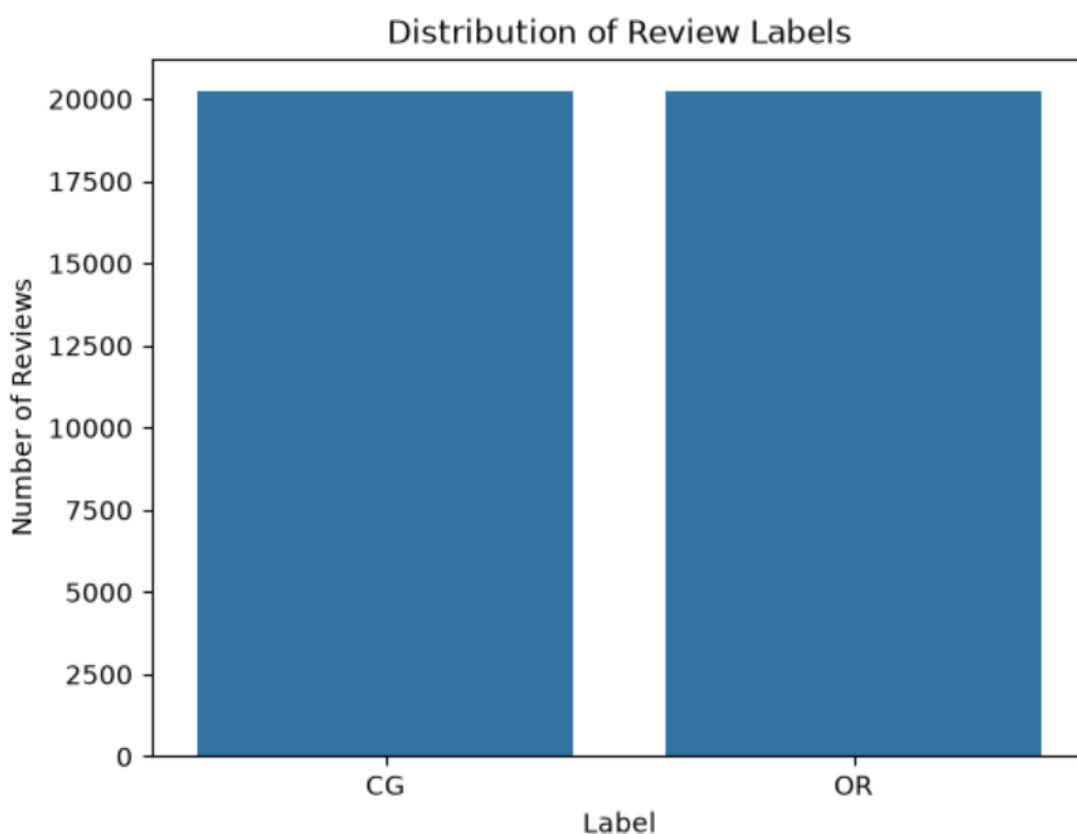


Figure 1. Distribution of Review Labels

The distribution confirms that both classes contain approximately the same number of samples, allowing model performance to be evaluated fairly without requiring additional balancing techniques.

The rating distribution was then examined to understand the characteristics of the reviews. As shown in **Figure 2**, most reviews received a rating of five stars, while lower ratings appear much less frequently. This pattern reflects the typical behavior observed in many online marketplaces, where users tend to submit positive feedback more often than negative reviews.

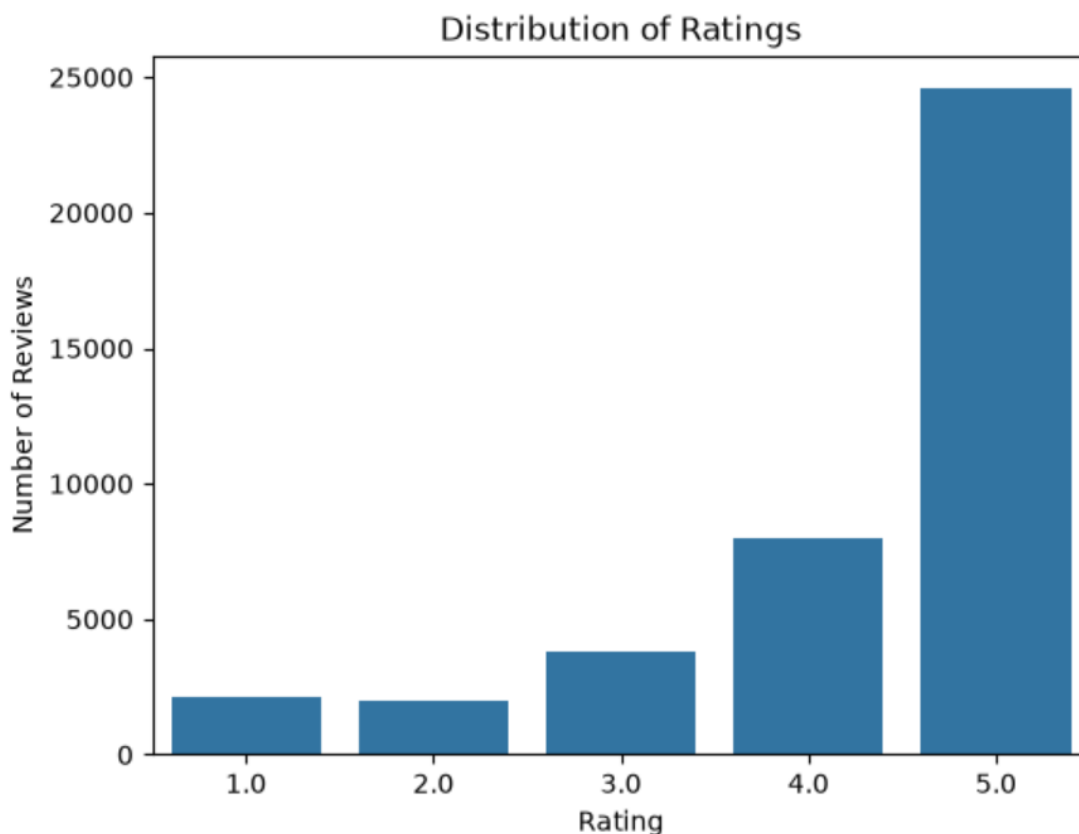


Figure 2. Distribution of Ratings

Next, the distribution of reviews across different product categories was analyzed. **Figure 3** shows that the dataset includes reviews from multiple Amazon categories such as books, electronics, home and kitchen, sports equipment, movies, toys, and pet supplies. The relatively balanced distribution across categories increases the diversity of the dataset and improves the model's ability to generalize.

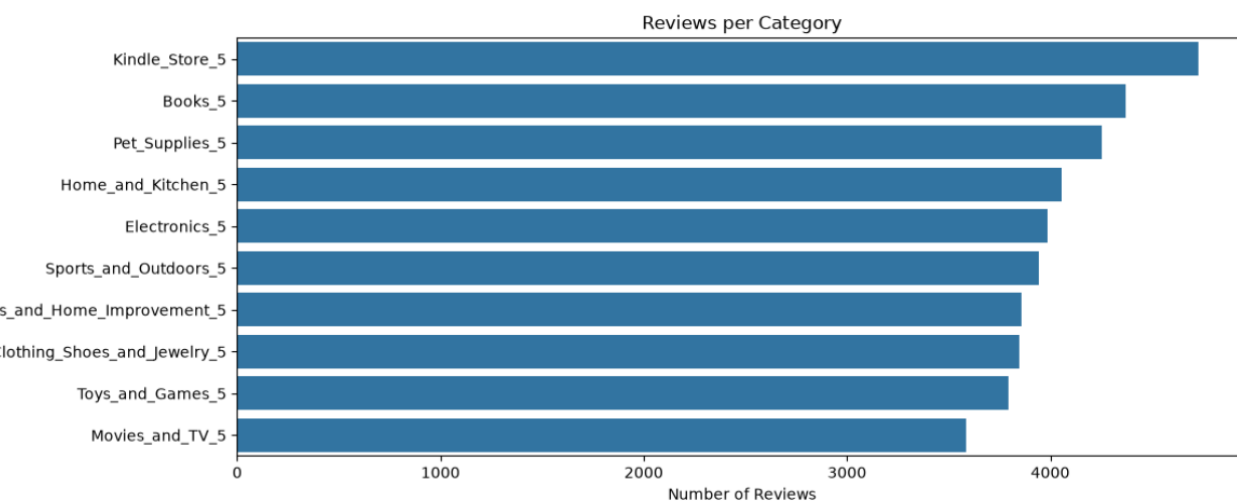


Figure 3. Reviews per Category

Review length was also investigated because fake and real reviews may differ in the amount of information they contain. As illustrated in **Figure 4**, real reviews generally contain more characters than fake reviews, although both classes exhibit a wide range of review lengths.

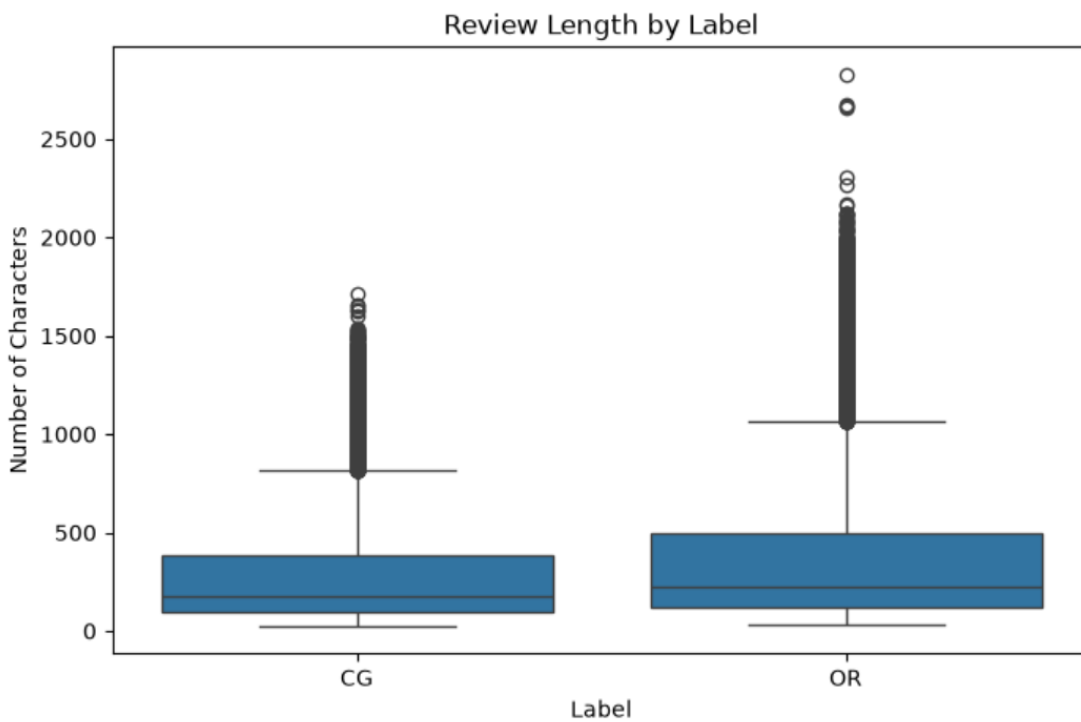


Figure 4. Review Length by Label

A similar pattern can be observed when comparing the number of words used in each review. **Figure 5** indicates that authentic reviews generally contain more words than fake reviews, suggesting that genuine customer experiences tend to provide more detailed descriptions.

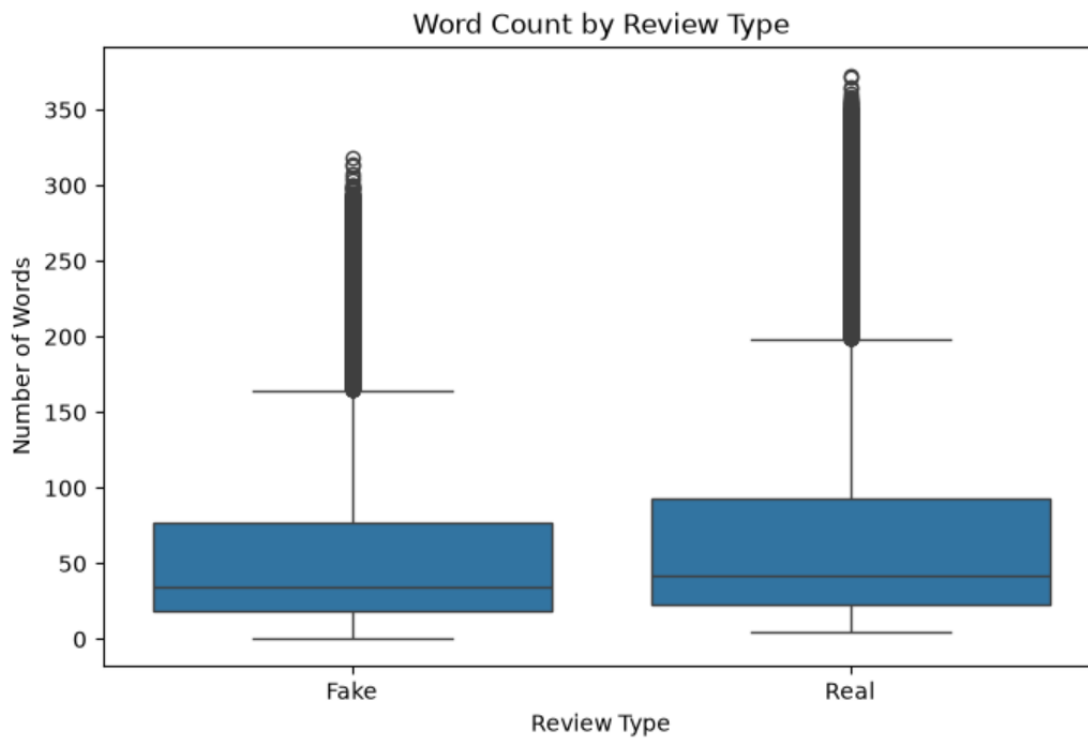


Figure 5. Word Count by Review Type

Finally, Word Clouds were generated for both fake and real reviews in order to visualize the most frequently occurring words. Although many common terms appear in both classes, the visualization provides an intuitive understanding of the vocabulary used within each review type and highlights the overall similarity of the language used by fake and authentic reviews.



Figure 6. Most Common Words in Fake Reviews



Figure 7. Most Common Words in Real Reviews

Overall, the exploratory analysis provided valuable insights into the dataset and confirmed that it is suitable for supervised machine learning. The balanced class distribution and the diversity of review categories make it an appropriate dataset for evaluating different text classification algorithms.

3 Methodology

The objective of this project was to build a machine learning model capable of classifying product reviews as either **Real** or **Fake**. To achieve this goal, a complete text classification pipeline was developed and several machine learning models were evaluated.

The first stage involved preprocessing the textual data. Since machine learning algorithms cannot process raw text directly, the reviews were converted into numerical feature vectors using the **Term Frequency–Inverse Document Frequency (TF-IDF)** representation. This technique assigns higher importance to words that frequently appear in a specific review while reducing the influence of extremely common words that appear in many reviews.

The dataset was divided into training and testing sets using an **80/20 split**. The training set was used to build the models, while the testing set was reserved for evaluating their performance on previously unseen reviews.

Several classical machine learning algorithms were trained and compared throughout the project. The development process began with **Multinomial Naive Bayes**, which served as the baseline model due to its simplicity and efficiency for text classification tasks. Next, **Logistic Regression** was evaluated, followed by a **Linear Support Vector Machine (Linear SVM)**, which demonstrated a noticeable improvement in classification performance.

To further improve the results, several enhancements were introduced. First, **bigram features** were added to the TF-IDF representation, allowing the model to learn meaningful two-word expressions instead of considering only individual words. Next, **GridSearchCV** was applied to optimize the model hyperparameters and identify the

best-performing configuration. Finally, the selected Linear SVM model was calibrated using **CalibratedClassifierCV**, enabling the model to produce probability estimates and confidence scores for each prediction displayed in the Streamlit application.

The complete workflow is summarized below:

1. Load the dataset.
2. Perform exploratory data analysis.
3. Preprocess the text.
4. Convert reviews into TF-IDF feature vectors.
5. Train multiple machine learning models.
6. Evaluate model performance.
7. Improve the best-performing model using feature engineering and hyperparameter tuning.
8. Deploy the final calibrated model as a Streamlit web application.

4 Results

Several machine learning models were evaluated throughout the project in order to identify the most effective approach for fake review detection. Each model was trained using the same training dataset and evaluated on the same testing dataset, allowing a fair comparison of their performance.

The evaluated models included:

- Multinomial Naive Bayes
- Logistic Regression
- Linear Support Vector Machine (Linear SVM)
- Linear SVM with Bigram Features
- Linear SVM with GridSearchCV
- Final Calibrated Linear SVM

The classification accuracy obtained by each model is presented in **Table 1**.

	Model	Accuracy
5	Final Model (Calibrated SVM)	0.9017
4	Linear SVM + GridSearch	0.9016
3	Linear SVM + Bigrams	0.8995
2	Linear SVM	0.8719
1	Logistic Regression	0.8660
0	Naive Bayes	0.8405

Table 1. Accuracy comparison of all evaluated models.

The results demonstrate a consistent improvement throughout the model development process. The baseline Multinomial Naive Bayes classifier achieved an accuracy of approximately **84%**. Replacing it with Logistic Regression and subsequently Linear SVM significantly improved the classification performance.

Introducing bigram features provided another noticeable improvement by allowing the model to capture meaningful word pairs instead of relying only on individual words. Hyperparameter optimization using GridSearchCV resulted in a small additional increase in accuracy.

Finally, probability calibration was applied to the best-performing Linear SVM model. Although the improvement in accuracy was relatively small, calibration enabled the model to generate reliable confidence scores for each prediction, making the final application more informative for end users.

Model Comparison

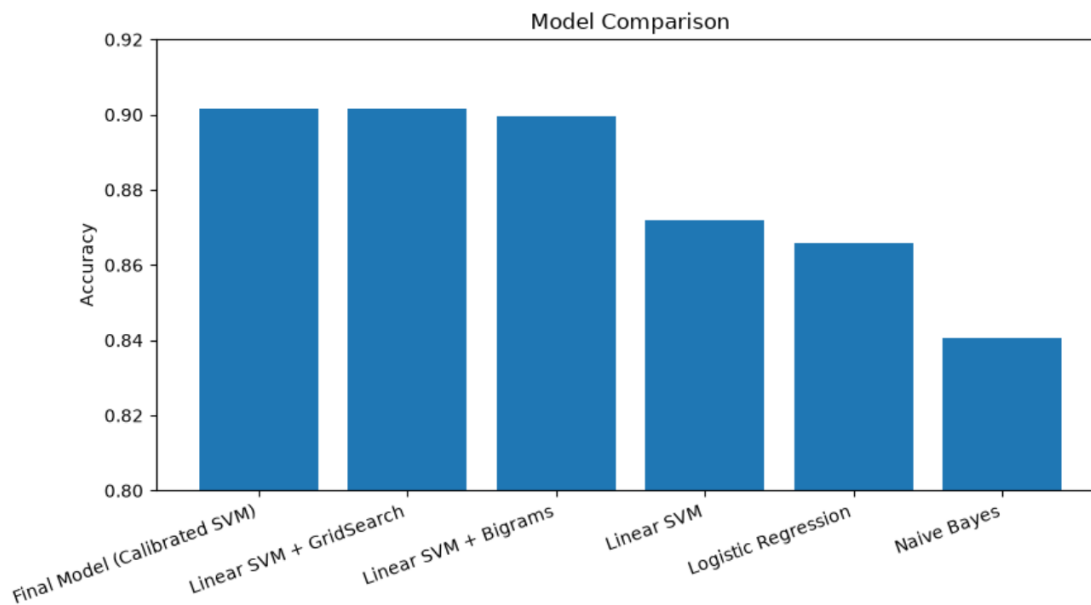


Figure 8. Accuracy comparison between all evaluated machine learning models.

As illustrated in **Figure 8**, the calibrated Linear SVM achieved the highest overall accuracy of approximately **90.17%**, outperforming all other evaluated models. For this reason, it was selected as the final model and deployed within the Streamlit application.

5 Streamlit Application

To make the final model easily accessible, a web application was developed using the **Streamlit** framework. The application provides a simple and user-friendly interface that allows users to analyze product reviews without requiring any programming knowledge.

The user simply enters a product review into the text box and clicks the **Analyze** button. The trained machine learning model then predicts whether the review is more likely to be **Real** or **Fake**. In addition to the predicted label, the application also displays a confidence score representing the model's certainty in its prediction.

The application loads the final calibrated Linear SVM model, ensuring that the displayed confidence values are meaningful probability estimates rather than raw classification scores. This makes the application more informative and practical for real-world use.

The interface was intentionally designed to remain clean and simple, focusing on ease of use while presenting only the most relevant information to the user.

Fake Review Detection

Enter a product review and the model will predict whether it is likely real or fake.

Review

I bought this water bottle about a month ago and have been using it every day. It keeps my drinks cold for hours, the lid closes tightly, and it feels durable. The only minor issue is that it is a bit heavier than I expected, but overall I am very satisfied with the purchase.

Analyze

Prediction: Real Review

Confidence: 94.28%

Figure 9. Streamlit application demonstrating a sample review prediction.

The final application successfully combines the trained machine learning model with an intuitive graphical interface, allowing users to classify product reviews within seconds.

6 Conclusion

This project demonstrated the complete development process of a machine learning solution for fake review detection. Starting from exploratory data analysis, several classical machine learning models were trained, evaluated, and compared using the same dataset.

Among all evaluated models, the calibrated Linear SVM achieved the highest performance with an accuracy of approximately **90.17%** while also providing reliable confidence scores for each prediction. The model was successfully integrated into a Streamlit application, making it easy to use through a simple graphical interface.

Overall, the project illustrates how traditional Natural Language Processing techniques combined with classical machine learning algorithms can achieve strong performance in text classification tasks without requiring computationally expensive deep learning models.

Future improvements could include evaluating transformer-based language models such as BERT, experimenting with larger datasets collected from multiple e-commerce platforms, and extending the system to support multilingual fake review detection.